LAB 1: AI-ASSISTED CITATION INTEGRITY AUDIT

Generate a Research Paper with AI, Then Audit Its Citations

AI Tools for Research • 2 Hours • Individual Activity • Topic IDs T1-T40

Student Detail	Fill in
Name	Abhishek
Roll Number	RSI2026507
Programme / Department	Ph.D.(IT)
Topic ID	37 ___
Full Assigned Topic	Retrieval Quality as the Primary Failure Point in RAG-Based and Agentic Research Assistants
AI Tool Used	
Date	21/08/2026

IMPORTANT: You will receive a topic from T1-T40. Your task is not to produce the "best" paper. Your task is to generate an AI-written research paper and systematically determine whether its citations can be trusted.

1. PURPOSE OF THIS LAB

Generative AI systems can produce research papers that look scholarly and may contain convincing references, author names, journal titles, DOIs, arXiv IDs, and in-text citations. However, the presence of a citation does not guarantee that the cited publication actually exists, that its metadata are correct, that the identifier belongs to that publication, or that the cited paper supports the statement made by the AI.

In this laboratory activity, you will test these issues yourself by generating a research paper and auditing a systematic sample of its references.

Question 1 - Citation Authenticity: Does the cited publication actually exist, and is it correctly identified?

Question 2 - Claim-Citation Support: If the publication is genuine, does it actually support the claim for which the AI cited it?

2. IMPORTANT RULES

- Use only the topic assigned to you by the instructor.
- Generate the paper in a fresh AI conversation.
- Do not tell the AI that you are conducting a citation audit or hallucination test.
- Do not intentionally ask the AI to generate fake references.
- Preserve the original AI-generated paper exactly as produced.
- Do not correct, regenerate, replace, or question references before saving the original output.
- For verification, do not accept another AI chatbot's statement as proof that a reference exists.
- Use scholarly databases, publisher pages, DOI/arXiv/PMID records, or other reliable scholarly sources.
- Record what you actually find, even if all references are genuine or all are problematic.

Most important principle: You are not being evaluated on whether the AI produces good or bad references. You are being evaluated on how systematically and correctly you audit them.

3. WHAT YOU WILL DO

Part	Activity
A	Record your baseline perception of AI citations
B	Record the AI environment used
C	Generate and preserve an AI-written research paper
D	Create a reference inventory
E	Select 10 references systematically
F	Judge their plausibility before verification
G	Verify authenticity, metadata and identifiers
H	Determine whether genuine references support the AI-generated claims

4. RECOMMENDED 2-HOUR TIME PLAN

Time	Activity
0-10 min	Parts A-B: Baseline and AI environment
10-30 min	Part C: Generate and save research paper
30-40 min	Parts D-E: Inventory and select references
40-45 min	Part F: Pre-verification plausibility
45-80 min	Part G: Authenticity and metadata audit
80-110 min	Part H: Claim-citation entailment audit
110-120 min	Calculate scores, check worksheet and prepare submission

Do not spend excessive time formatting the AI-generated paper. The audit is the main activity.

PART A - BASELINE ASSESSMENT

Complete this section BEFORE opening the AI tool.

A1. How often do you use generative AI for academic work?

☐ Never ☐ Rarely ☐ Occasionally ☐ Weekly ☐ Almost daily

A2. Have you previously asked an AI system to generate references?

☐ Frequently ☐ Occasionally ☐ Once or twice ☐ Never

A3. Have you previously checked whether an AI-generated citation actually exists?

☐ Always ☐ Sometimes ☐ Rarely ☐ Never ☐ Not applicable

A4. Before this lab, how confident are you that citations generated by AI are generally reliable?

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 (1 = not at all confident; 5 = completely confident)

Important: Do not change this answer later, even if your opinion changes during the experiment.

PART B - RECORD YOUR AI ENVIRONMENT

Information	Your Response
AI tool used	Chatgpt
Model name	Not Shown
Model/version shown	Not Shown

Access tier	☐ Free ☐ Paid/Pro ☐ Institutional ☐ Not sure
Web browsing available?	☐ Yes ☐ No ☐ Not sure
Was web browsing actually used?	☐ Yes ☐ No ☐ Not sure
Research/Deep Research mode used?	☐ Yes ☐ No
Date/time generated	21/08/2022

Important: If the model/version is not displayed, write NOT SHOWN. Do not guess the model version.

PART C - GENERATE THE AI RESEARCH PAPER

C1. Open a Fresh AI Conversation

Do not tell the AI that this is a citation audit, that you are testing hallucinations, or that you expect fabricated references. We want to observe normal AI research-paper generation behaviour.

C2. Use the Following Common Prompt

Replace only [Retrieval Quality as the Primary Failure Point in RAG-Based and Agentic Research Assistants].

> *Generate a scholarly research paper on the topic: "Retrieval Quality as the Primary Failure Point in RAG-Based and Agentic Research Assistants". The paper should be suitable for an academic audience and should not exceed approximately 10 pages. Include Title, Abstract, Keywords, Introduction, Background/Related Work/Literature Review, Main Thematic or Technical Discussion, Current Approaches or Developments, Research Challenges and Limitations, Research Gaps and Future Directions, Conclusion, and Complete References. Use scholarly literature to support factual, technical, comparative and research claims. Provide in-text citations throughout the paper. For every reference, provide as much bibliographic information as available, including authors, title, publication year, journal/conference/source, and DOI, PMID, arXiv ID or another persistent identifier wherever available. Use genuine scholarly sources. Do not intentionally fabricate references.*

C3. Preserve the Original Output

☐ Save the complete AI-generated paper

☐ Save the complete reference list

☐ Take screenshot(s) showing the generated output/references

☐ Save conversation/export/share evidence if available

STOP: Do not ask the AI to correct or verify anything yet. Your original output is the experimental evidence.

PART D - PAPER PROFILE AND REFERENCE INVENTORY

Paper title: .Retrieval Quality as the Primary Failure Point in RAG-Based and Agentic Research Assistants

Measure	Value
Approximate pages	12
Approximate word count	**5,321 words**
Number of sections	10 main sections
Total number of references	22
Approximate in-text citation occurrences	45
Did AI warn you to verify references?	☐ Clearly ☐ Vaguely ☐ No

D1. Reference Inventory

Classify every reference once using the best available identifier in this priority order: DOI -> PMID -> arXiv ID -> URL only -> No persistent identifier. The category totals must equal the total number of references in the AI-generated paper.

Reference Type	Count
DOI available	10

PMID available, no DOI	0
arXiv ID available, no DOI/PMID	10
URL only	0
No persistent identifier supplied	2
TOTAL	22

Examples

Category	Example	How to Count
DOI available	Dense Passage Retrieval for Open-Domain Question Answering." Proceedings of EMNLP 2020, pp. 6769–6781. DOI: 10.18653/v1/2020.emnlp-main.550. ,. "Lost in the Middle: How Language Models Use Long Contexts." Transactions of the Association for Computational Linguistics, 12, 157–173. DOI: 10.1162/tacl_a_00638. "Query2doc: Query Expansion with Large Language Models." Proceedings of EMNLP 2023, pp. 9414–9423. DOI: 10.18653/v1/2023. ,"RAGAs: Automated Evaluation of Retrieval Augmented Generation." Proceedings of EACL 2024: System Demonstrations, pp. 150–158. DOI: 10.18653/v1/2024 ,"ARES: An Automated Evaluation Framework for Retrieval-Augmented Generation Systems." Proceedings of NAACL 2024, pp. 338–354. DOI: 10.18653/v1/2024. ,"RAGTruth: A Hallucination Corpus for Developing Trustworthy Retrieval-Augmented Language Models." Proceedings of ACL 2024, pp. 10862–10878. DOI: 10.18653/v1/2024. ,"Assisting in Writing Wikipedia-like Articles From Scratch with Large Language Models." Proceedings of NAACL 2024, pp. 6252–6278. DOI: 10.18653/v1/2024. ,"Search-o1: Agentic Search-Enhanced Large Reasoning Models." Proceedings of EMNLP 2025, pp. 5420–5438. DOI: 10.18653/v1/2025. ,"DeepResearcher: Scaling Deep Research via Reinforcement Learning in Real-world Environments." Proceedings of EMNLP 2025, pp. 414–431. DOI: 10.18653/v1/2025. ,"Agent Laboratory: Using LLM Agents as Research Assistants." Findings of EMNLP 2025, pp. 5977–6043. DOI: 10.18653/v1/2025. ,"WebVoyager: Building an End-to-End Web Agent with Large Multimodal Models." Proceedings of ACL 2024, pp. 6864–6890. DOI: 10.18653/v1/2024. ,"Retrieval Augmented Generation or Long-Context LLMs? A Comprehensive Study and Hybrid Approach." Proceedings of EMNLP 2024: Industry Track, pp. 881–893. DOI: 10.18653/v1/2024.	DOI available
PMID available, no DOI	0	PMID available, no DOI
arXiv ID available	Yao, S., et al. ReAct: Synergizing Reasoning and Acting in Language Models. arXiv:2210.03629 Lewis, Patrick; Perez, Ethan; Piktus, Aleksandra; Petroni, Fabio; Karpukhin, Vladimir; Goyal, Naman; Küttler, Heinrich; Lewis, Mike; Yih, Wen-tau; Rocktäschel, Tim; Riedel, Sebastian; Kiela, Douwe. (2020). arXiv:2005.11401. Thakur, Nandan; Reimers, Nils; Rücklé, Andreas; Srivastava, Abhishek; Gurevych, Iryna. (2021) . arXiv:2104.08663 Gao, Luyu; Ma, Xueguang; Lin, Jimmy; Callan, Jamie. (2023). arXiv:2212.10496 Saad-Falcon, Jon; Khattab, Omar; Potts, Christopher; Zaharia, Matei. (2024). "ARES: An Automated Evaluation Framework for Retrieval-Augmented Generation Systems rXiv:2311.09476 Yan, Shi-Qi; Gu, Jia-Chen; Zhu, Yun; Ling, Zhen-Hua. (2024). "Corrective Retrieval Augmented Generation arXiv preprint. arXiv:2401.15884. Asai, Akari; Wu, Zeqiu; Wang, Yizhong; Sil, Avirup; Hajishirzi, Hannaneh. (2024). arXiv:2310.11511. Yu, Yue; Ping, Wei; Liu, Zihan; Wang, Boxin; You, Jiaxuan; Zhang, Chao; Shoeybi, Mohammad; Catanzaro, Bryan. (2024). arXiv:2407.02485. Deng, Xiang; Gu, Yu; Zheng, Boyuan; Chen, Shijie; Stevens, Samuel; Wang, Boshi; Sun, Huan; Su, Yu. (2023). arXiv:2306.06070	arXiv ID available

	Lin, Jimmy; Nogueira, Rodrigo; Yates, Andrew. (2020). "Pretrained Transformers for Text Ranking: BERT and Beyond." arXiv preprint. arXiv:2010.06467	
URL only		URL only
No persistent identifier	Zhou, Shuyan; Xu, Frank F.; Zhu, Hao; Zhou, Xuhui; Lo, Robert; Sridhar, Abishek; Cheng, Xianyi; Ou, Tianyue; Bisk, Yonatan; Fried, Daniel; Alon, Uri; Neubig, Graham. (2024). "WebArena: A Realistic Web Environment for Building Autonomous Agents."	No persistent identifier if none is supplied

Important: Absence of a DOI does not mean a reference is fake. An identifier being present also does not prove authenticity; it must be checked.

PART E - SELECT 10 REFERENCES FOR AUDIT

You are not allowed to choose only suspicious-looking references. This prevents selection bias.

If the bibliography contains 10 or fewer references: audit all references.

If it contains more than 10 references: select the first 3 references, the last 3 references, and 4 references approximately evenly distributed through the middle.

Example: If there are 28 references, a reasonable sample could be 1, 2, 3, 10, 12, 13,14, 20, 21, 22.

Audit Slot	Original Bibliography Reference No.
R1	1
R2	2
R3	3
R4	10
R5	12
R6	13
R7	14
R8	20
R9	21
R10	22

Important: From this point onward, R1-R10 refer to your audit sample, not necessarily bibliography references 1-10.

PART F - PLAUSIBILITY TEST

Do this BEFORE searching for any reference.

Look only at the citation as generated by the AI. Ask yourself: "If I had not verified this citation, how convincing would it look to me?"

Score	Meaning
1	Obviously suspicious
2	Somewhat suspicious
3	Unsure
4	Looks genuine
5	Completely convincing

Ref.	Initial Impression	Plausibility 1-5	Predict Genuine?
R1	Highly realistic foundational RAG paper; complete authors,	5	Yes

	venue, volume, pages and arXiv ID		
R2	Very convincing EMNLP citation with DOI, arXiv ID and page range	5	Yes
R3	Looks like a genuine benchmark paper; realistic NeurIPS D&B citation and arXiv ID	4	Yes
R4	Highly convincing journal citation with volume, pages and DOI	5	Yes
R5	Realistic ACL-style citation with authors, year and arXiv identifier	4	Yes
R6	Very convincing EMNLP citation with DOI, pages and arXiv ID	5	Yes
R7	Highly convincing EACL demonstration citation with DOI and pages	5	Yes
R8	Highly convincing NAACL citation with DOI, pages and arXiv ID	5	Yes
R9	Plausible arXiv citation, although less bibliographic detail is supplied	4	Yes
R10	Highly convincing ICLR citation with authors, year and arXiv ID	5	Yes

Example: A citation with realistic authors, a well-known journal, complete volume/pages and a DOI-like identifier may look highly convincing. You might record Plausibility = 5 and Predict Genuine = Yes. You are recording your pre-verification belief, not the truth.

Critical rule: Never change Part F after completing Part G. Your incorrect predictions are valuable experimental observations.

PART G - CITATION AUTHENTICITY AND METADATA AUDIT

G1. What Are You Checking?

- Does the publication exist?
- yes
- Is the title correct?
- yes
- Are the authors correct?
- yes
- Is the year correct?
- yes
- Is the journal/conference/source correct?
- yes
- Are volume, issue and pages correct, if supplied?
- Yes
- Does the DOI/arXiv/PMID belong to this publication?
- yes

G2. Verification Procedure

☐ Persistent identifier: DOI / arXiv / PMID

☐ Publisher or official repository

☐ Google Scholar

☐ Semantic Scholar / OpenAlex

☐ Crossref

☐ PubMed or another discipline-specific database

☐ Exact-title search

☐ Author + title-keyword search

Not acceptable as sole evidence: "I asked another AI and it said the paper exists." AI can help you search, but final verification must rely on scholarly or publisher evidence.

G3. Assign One A-E Code

Code	Classification	Meaning
A	VERIFIED	Publication exists and important metadata are correct.
B	WRONG METADATA	Publication exists, but one or more bibliographic details are incorrect.
C	FRANKENSTEIN	Real-looking pieces are combined, but that specific publication does not exist.
D	FABRICATED	No reliable scholarly evidence that the publication exists.
E	IDENTIFIER MISMATCH	DOI/arXiv/PMID is invalid or resolves to a different publication.

G4. Examples

A: AI gives a paper and all details match the publisher record.

B: The paper exists, but the AI gives the wrong publication year or journal.

C: The authors are real, the journal is real, and the topic is plausible, but no such paper with that title/authorship combination exists.

D: You cannot find the paper or a credible scholarly record after systematic searching.

E: The DOI supplied by AI opens an entirely different paper or does not resolve.

G5. Record Your Evidence

Ref.	Publication Found?	Title Match?	Authors Match?	Year Match?	Venue Match?	Identifier Match?
R1	Y	Y	Y	Y	Y	Y
R2	Y	Y	Y	Y	Y	Y
R3	Y	Y	Y	Y	NA	Y
R4	Y	Y	Y	Y	Y	Y
R5	Y	Y	Y	Y	Y	Y
R6	Y	Y	Y	Y	Y	Y
R7	Y	Y	Y	Y	Y	Y
R8	Y	Y	Y	Y	Y	Y
R9	Y	Y	Y	Y	Y	Y
R10	Y	Y	Y	Y	Y	Y

Ref.	Verification Source(s)	Identifier Checked	What You Found
R1			
R2			
R3			
R4			
R5			
R6			
R7			
R8			
R9			
R10			

G6. Final Classification

Ref.	Final Code	One-Line Reason
R1	A	Publication exists and the supplied title, authors, year, venue, pages and arXiv identifier match.
R2	A	Publication exists and the supplied EMNLP metadata and DOI/arXiv identifiers match
R3	A	Publication exists in the NeurIPS Datasets and Benchmarks track and the supplied bibliographic details match; no persistent identifier was supplied in the citation
R4	A	Publication exists in TACL and the title, authors, year, volume, pages and DOI match.
R5	A	Publication exists and the title, authors, year and arXiv identifier match.

R6	A	Publication exists and the EMNLP metadata, DOI, pages and arXiv identifier match.
R7	A	Publication exists and the EACL metadata, DOI and page range match
R8	A	Publication exists and the NAACL metadata, DOI and arXiv identifier match.
R9	A	Publication exists and the title, authors, year and arXiv identifier match.
R10	A	Publication exists and the title, authors, year and arXiv identifier match.

Good reason: B: Article exists, but AI reported 2022 instead of 2020.

Poor reason: B: Details wrong.

PART G - SCORING

Count your final classifications: A = 10____ B = __0__ C = __0__ D = _0___ E = __0__

Total audited: N = A + B + C + D + E = _10___

Code	Points
A	10
B	0
C	0
D	0
E	0

Authenticity Points = 4A + 3B + C + E

Authenticity Score = [(4A + 3B + C + E) / (4N)] x 100

My calculation: 4(__10__) + 3(0____) + 1(__0__) + 0(__0__) + 1(__0__) = _40___ points

Authenticity Score = ___40_ / (4 x _10___) x 100 = ____100__ /100

Worked example: If A=6, B=2, C=1, D=1, E=0, then points = 4(6)+3(2)+1(1)=31; maximum = 40; Authenticity Score = 31/40 x 100 = 77.5.

Score	Interpretation
90-100	Excellent authenticity
75-89	Good, minor concerns
60-74	Moderate citation-authenticity risk
40-59	High citation-authenticity risk
Below 40	Very high citation-authenticity risk

G7. Prediction Accuracy

Compare your Part F Genuine Yes/No predictions with the Part G results.

Prediction Accuracy = (Correct genuine/fake predictions / Total predictions) x 100

Correct predictions = __10__ Total predictions = __10__ Prediction Accuracy = __100____ %

FINAL LAB 1 RESULTS

Measure	Your Result
Total references in AI paper	21
References deeply audited	10
A: Verified	10
B: Wrong metadata	0
C: Frankenstein	0
D: Fabricated	0
E: Identifier mismatch	0
Authenticity Score	____100 /100
Prediction Accuracy	100%

FINAL INTERPRETATION

1. Did professional-looking citations necessarily turn out to be genuine?

In this audit sample, yes. All ten professional-looking citations were verified as genuine publications with matching bibliographic metadata. However, the audit also demonstrates why appearance alone is not sufficient: each citation still needs independent verification.

2. Did all genuine references actually support the claims for which they were cited?

Publication authenticity and claim support are different checks. The ten references were verified as genuine and their bibliographic metadata matched, but this audit does **not by itself establish that every citation fully supports every claim made in the paper**. Claim-level citation entailment would require a separate content-support audit.

3. What was the most serious citation-integrity problem you observed?

For R1–R10, **no serious citation-integrity problem was observed**. All ten were classified A (VERIFIED). The main potential risk identified by the audit procedure is that plausible-looking citations can appear trustworthy without verification, so metadata and identifiers must still be checked independently.

4. What is the single most important lesson you learned from this lab?

The most important lesson is that citation appearance is not evidence of authenticity. A citation should be verified against a scholarly or publisher record, checking the publication's existence, title, authors, year, venue, bibliographic details and persistent identifier before treating it as reliable.

SUBMISSION INSTRUCTIONS

Submit one PDF: Lab1_RollNumber_Name

Example: Lab1_IEC2026001_RahulSharma

STUDENT DECLARATION

I confirm that I generated the research paper using the recorded AI environment, preserved the original output before conducting the audit, followed the prescribed reference-selection procedure, and personally verified the evidence reported in this worksheet. I have not altered my pre-verification predictions after seeing the verification results.

Student Name: .....Abhishek.....

Roll Number: .....RSI2026507.....

Signature: .....

Date: .....21/08/2026.....

CENTRAL LESSON OF LAB 1

Does the source exist? + Does the source actually support the claim?

AI Generation -> Reference Inventory -> Systematic Sampling -> Prediction -> Authenticity Verification -> Claim-Support Verification -> Quantitative Scores -> Submission